DEEP LEARNING ASSESSMENT · FINAL REPORT

Energy Consumption Prediction

Multivariate Appliance Energy Prediction Dataset

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1. Introduction

Energy consumption forecasting is a critical component of smart building management and sustainable energy systems. This report presents a comprehensive study on predicting appliance energy consumption using the **Multivariate Appliance Energy Prediction Dataset**, a well-known benchmark containing high-resolution sensor measurements collected from a low-energy house in Belgium.

The primary objective is to develop, evaluate, and compare machine learning and deep learning models that can accurately predict the energy consumed by home appliances. The pipeline encompasses exploratory data analysis, robust preprocessing, advanced feature engineering, model training using Random Forest as a baseline and multiple LSTM architectures as the deep learning component, and rigorous performance evaluation.

The dataset contains **19,735 observations** recorded at **10-minute intervals** spanning January–May 2016. The target variable, *Appliances*, represents energy consumption in Wh, while 28 features capture indoor/outdoor temperature, humidity, atmospheric pressure, wind speed, visibility, dew point, and light usage.

2. Data Insights — Exploratory Data Analysis

2.1 Dataset Overview

The raw dataset consists of **19,735 rows × 29 columns**. All variables are numeric (26 float64, 2 int64) except the timestamp column (object). No missing values were found.

Property	Value
Total Records	19,735
Total Features	29 (including target)
Data Types	float64 (26), int64 (2), object (1)
Time Resolution	10-minute intervals
Date Range	January – May 2016
Missing Values	0 (none)
Target Variable	Appliances (Wh energy consumption)

2.2 Correlation Analysis

A full correlation heatmap revealed two key structural characteristics of the dataset:

- The target variable **Appliances** exhibits **low correlation** with all features. The highest Pearson |r| value is 0.197 (lights), indicating that energy consumption is not linearly predictable from individual sensor readings alone.
- Significant **multicollinearity** exists among the temperature (T1–T9, T_out) and humidity (RH_1–RH_9, RH_out) groups, as expected from spatially correlated sensors within the same building.

Feature	Correlation with Appliances (|r|)
lights	0.1973

RH_out	0.1523
T2	0.1201
T6	0.1176
T_out	0.0992
RH_8	0.0940
Windspeed	0.0871
rv1 / rv2	0.0111 (random noise variables)
Visibility	0.0002 (lowest)

2.3 Temporal Patterns

Analysing energy consumption across different time dimensions revealed meaningful patterns:

- **Hourly pattern:** Appliance energy peaks around **6:00 PM**, consistent with typical residential occupancy. Consumption is lowest in the early morning hours (midnight to ~6:00 AM).
- **Daily pattern: Sunday** records the highest mean energy consumption while **Monday** records the lowest, likely reflecting weekend vs. weekday activity.
- **Seasonal decomposition** (weekly period) confirmed a stable seasonal cycle with no long-term upward or downward trend over the observation window.
- **Monthly pattern:** February had the highest mean consumption (100.95 Wh) and May the lowest (94.20 Wh); the variation is moderate.

2.4 Feature Distributions

Histograms of all variables showed that most features are approximately normally distributed. A right-skewed distribution is evident for the Appliances target — with most readings clustered in low energy ranges and a long tail of high-consumption events. The random variables rv1 and rv2 share identical distributions, confirming they are symmetric noise injected into the dataset as control variables.

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3. Preprocessing

3.1 Missing Value Handling

A complete check across all 29 columns confirmed **zero missing values**. No imputation was required at this stage.

3.2 Outlier Detection & Imputation

Outliers in the Appliances column were identified using the **Interquartile Range (IQR) method**:

- Q1 = 25th percentile, Q3 = 75th percentile, IQR = Q3 – Q1
- Lower fence = Q1 – 1.5 × IQR; Upper fence = Q3 + 1.5 × IQR

Metric	Value
Outliers Detected (IQR method)	2,138 records
Percentage of Dataset	10.8%

Max Mean Outlier Day	April 18, 2016 (463.3 Wh avg)
Peak Outlier Hour	11 PM (380 Wh avg)
Decision	Imputation (not removal)

Because the outlier analysis showed that high-consumption events are distributed across all hours and months without a dominant erroneous pattern, they were deemed **genuine real-world events** rather than data-entry errors. Removal would have deleted valuable peak-demand information.

A **two-stage imputation pipeline** was applied:

- **Stage 1 – Linear Interpolation:** Outliers were set to NaN and interpolated linearly from adjacent valid readings. While this removed most extreme values, some residual outliers remained.
- **Stage 2 – Rolling Window Mean (window=15):** Remaining outliers were replaced with a centred 15-period rolling mean, fully eliminating all extreme values while preserving the temporal structure of the series.

Note: The cleaned column was saved as Appliances_imputed and used for all subsequent analysis.

3.3 Stationarity Analysis

Both the **Augmented Dickey-Fuller (ADF)** and **KPSS** tests were applied to the Appliances series. The ADF test rejected the unit-root null hypothesis, and the KPSS p-value (≥ 0.1) indicated stationarity. The Appliances series is therefore **stationary**, which simplifies modelling and validates the use of lag features without differencing.

4. Feature Engineering

4.1 Lag Features — ACF Analysis

The **Autocorrelation Function (ACF)** plot for Appliances_imputed with 60 lags was examined. Statistically significant autocorrelation was observed up to **lag 34**. Accordingly, 34 lagged copies of the target variable (lag_1 … lag_34) were created and appended as features.

Using lagged values of the target (rather than the target itself at time t) is critical to **prevent data leakage** — the model only sees past observations when predicting the current value.

4.2 Rolling Window Statistics

To capture recent trend and volatility, four rolling statistics were computed on Appliances_imputed, all shifted by one step to avoid leakage:

Feature	Window	Statistic	Rationale
roll_mean_1hour	6 steps (1 h)	Mean	Short-term trend level
roll_std_1hour	6 steps (1 h)	Std	Short-term volatility
roll_mean_3hour	18 steps (3 h)	Mean	Medium-term trend level
Feature	**Window**	**Statistic**	**Rationale**
roll_std_3hour	18 steps (3 h)	Std	Medium-term volatility

4.3 Interaction Features (Temperature × Humidity)

Domain knowledge suggests that the **combined effect** of temperature and humidity on human comfort — and therefore on HVAC/appliance usage — is more informative than either variable alone. All pairwise products of the 10 temperature sensors and 10 humidity sensors were computed, yielding **100 interaction terms** (T1_x_RH_1, T1_x_RH_2, …, T_out_x_RH_out).

After all feature engineering steps and dropping rows with NaN values (from lag/rolling windows), the final dataset contains approximately **19,663 rows**. For the LSTM model, a correlation-based selection (|r| > 0.1 with target) combined with a multicollinearity filter (dropping features with pairwise |r| > 0.8) reduced the feature space to **13 input features**.

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5. Model Design

5.1 Baseline — Random Forest Regressor

A **Random Forest Regressor** was used as the baseline model to benchmark performance before moving to deep learning. The dataset was split chronologically (80/20) to respect temporal ordering, then standardised with StandardScaler (fit on training only).

Hyperparameter choices for the baseline RF model:

Hyperparameter	Value	Rationale
n_estimators	200	Sufficient ensemble diversity without excessive compute
max_depth	10	Moderate depth to balance bias/variance
min_samples_split	15	Regularisation to avoid overfitting
min_samples_leaf	8	Further smoothing of leaf predictions
max_features	0.3	30% of features per split — reduces correlation
max_samples	0.4	Bagging with 40% of data — diversity + speed

Feature selection for RF: Two strategies were explored — Permutation Importance (selecting features with positive mean importance) and Correlation Analysis (|r| > 0.1 with target, then removing multicollinear pairs at threshold 0.9). The correlation-based model was used as the primary RF benchmark.

5.2 Deep Learning — LSTM Models

Long Short-Term Memory (LSTM) networks were chosen for their proven ability to capture long-range temporal dependencies in sequential data. The energy consumption series has strong autocorrelation up to 34 lags, making LSTM architectures well-suited.

Input features were scaled with **MinMaxScaler** (fit on training set only). Each row was reshaped from (n_features,) to (n_features, 1) using a Reshape layer, treating each feature as a time step in the LSTM sequence.

Six LSTM architectures (Model 1 – Model 6) were trained and evaluated:

Model	Architecture	Optimizer	Batch	Activation
Model 1	LSTM(32)→LSTM(16)→LSTM(8)→Dense(1)	Adam	5	Default (tanh) + L2
Model 2	LSTM(32)→LSTM(16)→LSTM(8)→Dense(1)	Adam	16	Default (tanh)

Model 3	LSTM(32)→LSTM(16)→LSTM(8)→Dense(1)	RMSprop	16	ReLU + Dense(relu)
Model 4	LSTM(32)→LSTM(16)→LSTM(8)→Dense(1)*	RMSprop	16	ReLU + Dense(relu)
Model 5	STM(32)→LSTM(16)→LSTM(8)→LSTM(4)→Dense(.)	RMSprop	16	Default+Dense(relu)
Model 6	LSTM(64)→LSTM(32)→LSTM(16)→Dense(1)	RMSprop	16	Default (tanh)

All LSTM models shared the following training configuration:

- Loss function: **Mean Squared Error (MSE)** — standard for regression.
- Metrics tracked: **MAE** alongside MSE for interpretability.
- Train/Validation split: 80% training, 20% held-out validation (chronological).
- Maximum epochs: 25; **EarlyStopping** (patience=15) to prevent overfitting.
- **ModelCheckpoint**: saves the best validation loss weights automatically.
- **ReduceLROnPlateau**: halves learning rate when validation loss plateaus (factor=0.5, patience=7, min_lr=1e-6).
- Dropout: 30% applied after each LSTM layer to regularise.
- GPU-accelerated training (NVIDIA GPU detected and used via TensorFlow device context).

6. Results

6.1 Random Forest Performance

The baseline RF model (correlation-based feature selection) achieved the following:

Metric	Train Set	Test Set
MAE	0.3189	0.3315
MSE	0.2083	0.2148
RMSE	0.4564	0.4635
R^2	0.7917	0.7370
MAPE (%)	65.66%	75.95%

6.2 LSTM Model Comparison

All six LSTM models outperformed the Random Forest baseline on the test set. The comparison table below summarises test-set MSE and MAE for each architecture:

Model	Architecture Summary	Test MSE	Test MAE	Rank
Model 1	3×LSTM(32,16,8) Adam L2 bs=5	0.01095	0.07268	2
Model	**Architecture Summary**	**Test MSE**	**Test MAE**	**Rank**
Model 2	3×LSTM(32,16,8) Adam bs=16	0.01238	0.07931	4
Model 3	3×LSTM(32,16,8) RMSp ReLU bs=16	0.01077	0.07520	3
Model 4	4×LSTM(32,16,8,4) RMSp ReLU bs=16	0.01111	0.07552	5 (tied)
Model 5	4×LSTM(32,16,8,4) RMSp Dense-ReLU	0.01111	0.07552	5 (tied)

Model 6	3×LSTM(64,32,16) RMSp bs=16	0.01028	0.07243	1 ★

Model 6 achieved the best overall performance: wider LSTM layers (64→32→16 units) with RMSprop optimiser delivered the lowest test MSE (0.01028) and lowest test MAE (0.07243). All LSTM models achieved substantially better results than the Random Forest baseline.

6.3 Best Model (Model 6) — Training Dynamics

Model 6 trained over the full 25 epochs without triggering early stopping, showing steady improvement. Selected epoch metrics:

Epoch	Train Loss	Train MAE	Val Loss	Val MAE
1	0.0408	0.1499	0.0266	0.1212
5	0.0263	0.1223	0.0207	0.1099
10	0.0158	0.0921	0.0124	0.0805
15	0.0141	0.0863	0.0111	0.0753
20	0.0135	0.0841	0.0106	0.0734
25	0.0131	0.0829	0.0104	0.0726
Test	—	—	0.01028	0.07243

The training and validation curves converged consistently without divergence, indicating good generalisation. The Actual vs. Predicted plot (visualised in the notebook) shows the LSTM correctly tracking the overall shape and magnitude of energy consumption on the test set.